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ORIGINAL ARTICLE

Coronary Artery Calcium Progression and Cognitive Function in Midlife—The CARDIA Study (Coronary Artery Risk Development in Young Adults)

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BACKGROUND: We aimed to examine how coronary artery calcium (CAC) and its progression relate to cognitive function in midlife, an important time for cognitive aging.

METHODS: We studied participants enrolled in the prospective CARDIA study (Coronary Artery Risk Development in Young Adults), a longitudinal cohort of Black and White adults aged 18 to 30 years at baseline, who completed CAC measurements using computed tomography at year 15 (2000–2001; our baseline), had at least 1 follow-up CAC measurement at years 20 (2005–2006) or 25 (2010–2011), and completed cognitive assessments with a battery of 5 tests at years 30 (2015–2016) or 35 (2020–2022). CAC progression was defined as: (1) CAC >0 at follow-up among participants with baseline CAC=0; (2) an annualized change of ≥10 units at follow-up among those with 0<baseline CAC <100; or (3) an annualized percent change ≥10% among those with baseline CAC ≥100. We used linear regression to estimate the cognitive performance (z-standardized) with 95% CIs of each cognitive test by baseline CAC score (0 versus >0) and CAC progression.

RESULTS: Among the 2341 participants (mean baseline age 40.3±3.6 years; 56% female), baseline CAC >0 (9%) was associated with lower processing speed, verbal memory, and global cognition, whereas CAC progression (26%) was associated with lower processing speed and global cognition after adjusting for demographics, education, physical activity, depressive symptoms, APOE ε4 allele, and baseline CAC score. The standardized cognitive differences (95% CI) for CAC progression versus no progression were −0.14 (95% CI, −0.23 to −0.06) for the digit symbol substitution test and −0.09 (95% CI, −0.17 to −0.01) for the Montreal cognitive assessment.

CONCLUSIONS: CAC progression was associated with worse midlife processing speed and global cognition, independent of baseline CAC score and established risk factors. Repeated CAC assessments may offer clinical value for identifying individuals at increased risk for midlife cognitive decline.

Key Words: atherosclerosis ■ demography ■ exercise ■ risk factors ■ tomography

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Cumulative exposure to cardiovascular disease (CVD) risk factors from early to middle adulthood has been associated with worse cognitive performance in midlife.[1] Coronary artery calcium (CAC), an established marker of subclinical atherosclerosis, can reflect the biological response to cumulative vascular injury and serve as a risk indicator of CVD, independent of conventional CVD risk factors.[2,3] Since CVD risk factors and subclinical atherosclerosis play an important role in cognitive aging, CAC may also be an important risk indicator. However, only a few prior studies have investigated the CAC score and cognitive function in midlife.[4–6]

Furthermore, progression of CAC over time may capture the dynamic nature of plaque development

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WHAT IS KNOWN

- Cumulative exposure to cardiovascular risk factors from early adulthood through midlife is associated with worse cognitive performance in midlife.
- Coronary artery calcium (CAC) is an established marker of subclinical atherosclerosis and reflects long-term vascular injury.
- Evidence on whether CAC and its progression are linked to cognitive performance in midlife has been limited and inconsistent.

WHAT THE STUDY ADDS

- CAC score and its progression over ≈10 years were associated with lower cognitive function in midlife, particularly processing speed and global cognition.
- The associations of CAC progression with worse cognitive performance remained after adjusting for baseline CAC, suggesting that CAC progression may be a risk indicator for early cognitive aging, independent of baseline CAC score and other risk factors.
- Assessment of CAC progression through repeated CAC scans may provide additional clinical value, beyond baseline CAC alone, for identifying individuals at increased risk for midlife cognitive decline.

Nonstandard Abbreviations and Acronyms

CAC	coronary artery calcium
CARDIA	Coronary Artery Risk Development in Young Adults
CVD	cardiovascular disease
DSST	digit symbol substitution test
MESA	Multi-Ethnic Study of Atherosclerosis
MoCA	Montreal cognitive assessment

better than a single measurement of CAC. Evidence is lacking as to whether CAC progression, determined by repeat CAC scans, is linked to lower cognitive performance.[7,8] To our knowledge, only 1 small study[7] of type 1 diabetes patients reported a positive association of CAC progression with cognitive decline. No study has examined the association in the general population or in early to mid-adulthood, although midlife is felt to be an important time for early intervention and prognosis of cognitive aging.[9]

Therefore, our goal was to examine the association of CAC score at baseline and CAC progression with cognitive performance in midlife as part of an ongoing prospective study. We hypothesized that CAC progression would be associated with worse cognitive performance in midlife, independent of baseline CAC and other risk factors.

METHODS

The CARDIA (Coronary Artery Risk Development in Young Adults) data are available on reasonable request from the CARDIA Coordinating Center.

The CARDIA study enrolled 5115 Black and White men and women aged 18 to 30 years from March 25, 1985, to June 7, 1986 (year 0) at 4 field centers.[10] Sampling was population-based and approximately balanced within the field center by sex, age, race/ethnicity, and education level. Written informed consent was obtained from study participants at each examination. Institutional review boards approved study protocols at each study site (Kaiser Permanente Division of Research, Oakland; Northwestern University, Feinberg School of Medicine, Department of Preventive Medicine; University of Minnesota, School of Public Health, Division of Epidemiology and Community Health; University of Alabama at Birmingham, School of Public Health, Department of Epidemiology) as well as at University of California, San Francisco. Of the 3671 CARDIA participants who returned to complete the year 15 follow-up exam (2000–2001), 2341 participants without a history of stroke (mean age 40.3±3.6 years, 56% female, 43% Black at baseline) completed a baseline CAC assessment and had at least 1 follow-up CAC measurement at years 20 (2005–2006) or 25 (2010–2011) and completed cognitive testing at years 30 (2015–2016) or 35 (2020–2022; Figure S1).

Measurement of CAC

At years 15, 20, and 25, CAC was measured as previously described.[11,12] Image analysts blinded to participant characteristics calculated a total CAC score using a modified Agatston method.[13] Based on the Agatston score at year 15, we divided the participants into 2 groups: CAC score >0 or =0. CAC progression was defined using CAC scores at years 15, 20, and 25 as (1) CAC >0 at follow-up among participants with baseline CAC=0; (2) an annualized change of ≥10 units at follow-up among those with 0< baseline CAC <100, calculated by subtracting the year 15 CAC score from the year 20 (dividing by 5 years) or the year 25 score (dividing by 10 years); or (3) an annualized percent change (annualized change in CAC score divided by baseline CAC score) ≥10% among those with baseline CAC ≥100.[14] The scanning protocol used in CARDIA, MESA (Multi-Ethnic Study of Atherosclerosis), and Jackson Heart Study is optimized for large epidemiological studies and reproducibility of CAC scoring using multiple CT systems has previously been described.[11,12,15] Interclass correlation coefficient was 0.999, and intra- and inter-reader error was 5.6% and 7.0% at year 25 for 156 scan pairs that readers reevaluated, blinded to previous reports.[16]

Assessment of Cognitive Function

Cognitive testing was conducted in English using a battery of 5 cognitive tests administered by CARDIA technicians at years 30 and 35. If both time points were available for a participant, we used the year 35 values, but used the year 30 measure if year 35 was not available (lower participation due to the COVID-19 pandemic). The digit symbol substitution test (DSST) assessed processing speed, executive function, and working memory (range, 0–133).[17] The Rey auditory verbal learning test measured verbal learning and memory, featuring a delayed score

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for words accurately recalled after a delay (range, 0–15).[18] The Stroop test measured executive function.[19] We used the negative of the Stroop test scores to make lower scores indicate poorer performance, to be consistent with other tests (range, −160 to 0). The Montreal cognitive assessment (MoCA) is a global cognitive function assessment (range, 0–30).[20] Fluency was determined by an average score from letter and category fluency tests (range ≥0).[21] All scores were z-standardized using the sample mean and SD, and lower scores indicate worse cognitive performance.

Covariates

At year 15, age, race, sex, years of education, current smoking, and current alcohol use were obtained from self-report and interviewer-administered questionnaires. We dichotomized the physical activity scale by the recommended level of physical activity (≥300 exercise units).[22] Depressive symptoms were assessed using the Center for Epidemiologic Studies Depression scale.[23] Hypertension was defined as a blood pressure of 140/90 mmHg or more, or use of antihypertensive medication. Dyslipidemia was defined as low-density lipoprotein cholesterol levels ≥160, non–high-density lipoprotein cholesterol levels ≥190, or the use of lipid-lowering medication. Body mass index ≥30 was defined as obesity. Diabetes was diagnosed as a fasting plasma glucose level ≥126 mg/dL or the use of medication for diabetes. The presence of the APOE ε4 allele was determined using standard methods.[24]

Statistical Analysis

We compared participant characteristics by baseline CAC score (>0 versus =0) and the progression of CAC, using the χ^2 test for categorical variables and Mann-Whitney *U* tests for continuous variables. Linear regression models were employed to estimate the marginal mean of cognitive tests by baseline CAC score, and the z-standardized cognitive differences were estimated for baseline CAC score and CAC progression. We tested that all assumptions of linear regression were met, and found no evidence of violation. Covariates were selected a priori based on prior literature and subject-matter knowledge.[4,5,25] Among the risk factors identified a priori, we sequentially adjusted for covariates to evaluate the incremental effects of each set. Model 1 was age-adjusted. Model 2 was additionally adjusted for sex, race, and education. Baseline CAC score was further included for analyses of CAC progression. Model 3, our primary model, further adjusted for midlife dementia risk factors other than conventional CVD risk factors, including physical activity, depressive symptoms, and APOE (no-ε4, ε4 heterozygotes, or ε4 homozygotes). In the exploratory model (model 4), we estimated the association between CAC progression and cognitive function by further adjusting for CVD risk factors (hypertension, diabetes, obesity, dyslipidemia, current smoking, and drinking), given the potential direct impacts of CVD risk factors on cognitive function independent of CAC pathways. Missing covariates were present in <2% of participants except for APOE ε4 status. APOE ε4 status was missing for 9% of the cohort, but was included in the model as a missing category. In sensitivity analyses, we repeated the analyses by re-categorizing baseline CAC scores using thresholds of 0, 1 to 99, and ≥100, which are commonly used in clinical practice.[26] Furthermore, we excluded participants with incident CVD (coronary heart disease, heart failure, stroke, and transient ischemic attack) before cognitive testing. Interactions by sex and race on the association between CAC progression and cognitive function were tested by adding a product term in the full adjustment model. All statistical tests were 2-sided, and $P<0.05$ was considered statistically significant. All analyses were performed using Stata Statistical Software: Release 18, College Station, Texas.

RESULTS

Among 2341 participants (mean baseline age, 40.3±3.6 years; mean age at cognitive testing, 59.0±4.6 years), 396 (16.9%) underwent a single follow-up CAC scan (167 [7.1%] at year 20 only and 229 [9.8%] at year 25 only), and 1945 (83.1%) underwent 2 follow-up scans at both time points. During an average follow-up of 9.6 years, CAC progression was observed in 619 (26.4%) participants.

The 211 (9.0%) participants with baseline CAC score >0 were older, more likely to be male, and had a higher prevalence of smoking, hypertension, and dyslipidemia compared with those with baseline CAC score=0 (Table 1). Compared with participants without CAC progression, participants with CAC progression were also older, more likely to be male, had less education, and had a higher prevalence of smoking, hypertension, dyslipidemia, and obesity.

Baseline CAC >0 was associated with lower marginal mean scores on all cognitive tests compared with CAC=0 (Table 2). After adjusting for demographics, education, physical activity, depressive symptoms, and APOE, the standardized cognitive differences for baseline CAC >0 remained statistically significant for DSST, Rey auditory verbal learning test, and MoCA but not for Stroop or fluency (model 3). Additional adjustment for other CVD risk factors did not materially change the association. Participants with CAC scores of 1 to 99 and ≥100 demonstrated similarly lower cognitive performance compared with those with a CAC score of 0, although the association for CAC ≥100 did not reach statistical significance (Table S1).

CAC progression was associated with a lower cognitive performance across all cognitive tests compared with those with no CAC progression in an age-adjusted model (Figure). After adjusting for other demographics, education, and baseline CAC score (model 2), the association with lower cognitive function remained significant for most of the cognitive tests (but not for the Rey auditory verbal learning test or Stroop).

In the primary model (model 3), with additional adjustment for physical activity, depressive symptoms, and APOE, the associations remained statistically significant for DSST and MoCA: the standardized cognitive differences (95% CI) for CAC progression versus no progression were −0.14 (95% CI, −0.23 to −0.06) for DSST and −0.09 (95% CI, −0.17 to −0.01) for MoCA. After

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Table 1. Baseline Characteristics of Participants in the CARDIA Study (Coronary Artery Risk Development in Young Adults) by Coronary Artery Calcium Score and Progression

	Baseline CAC score (year 15)			CAC progression†		
	0	>0		No progression	Progression	
Variable‡	(n=2130)	(n=211)	*P* value*	(n=1722)	(n=619)	*P* value*
Age, y	41 (37–43)	43 (40–44)	<0.001	40 (37–43)	42 (39–44)	<0.001
Female, n (%)	1247 (58.5)	59 (28.0)	<0.001	1076 (62.5)	230 (37.2)	<0.001
Black race, n (%)	941 (44.2)	71 (33.7)	0.003	757 (44.0)	255 (41.2)	0.23
Education, y	16 (13–16)	14 (12–16)	0.009	16 (13–16)	15 (13–16)	0.002
Depressive symptoms, n (%)	311 (14.7)	28 (13.5)	0.62	248 (14.5)	91 (14.9)	0.85
Current smoker, n (%)	379 (17.8)	57 (27.1)	0.001	283 (16.5)	153 (24.8)	<0.001
Current drinker, n (%)	1707 (80.2)	170 (81.0)	0.79	1382 (80.4)	495 (80.0)	0.84
Physical activity ≥300 exercise units, n (%)§	1018 (47.8)	123 (58.3)	0.004	832 (48.3)	309 (49.9)	0.5
Hypertension, n (%)	277 (13.0)	58 (27.5)	<0.001	186 (10.8)	149 (24.1)	<0.001
Dyslipidemia, n (%)	189 (9.0)	48 (23.5)	<0.001	134 (7.9)	103 (17.0)	<0.001
Diabetes, n (%)	80 (3.8)	17 (8.1)	0.003	64 (3.7)	33 (5.3)	0.08
Obesity, n (%)‖	669 (31.5)	82 (38.9)	0.03	502 (29.2)	249 (40.3)	<0.001
Presence of APOE-ε4, %	589 (30.5)	56 (29.8)	0.85	476 (30.5)	169 (30.1)	0.84

CAC indicates coronary artery calcium.

*Mann-Whitney *U* tests for continuous variables and χ^2 test for categorical variables.

†CAC progression is defined using CAC scores at years 15, 20, and 25 as (1) CAC >0 at follow-up among participants with baseline CAC=0; (2) an annualized change of ≥10 units at follow-up among those with 0< baseline CAC <100, calculated by subtracting the year 15 CAC score from the year 20 average (dividing by 5 y) or the year 25 score (dividing by 10 y); or (3) an annualized percent change (annualized change in CAC score divided by baseline CAC score) ≥10% among those with baseline CAC ≥100.

‡Data are median (interquartile range) for continuous variables and percentages for categorical variables. Percentages were calculated using the nonmissing denominator for each variable.

§Physical activity is summarized as exercise units using the CARDIA Physical Activity History Questionnaire. Physical activity ≥300 exercise units indicates achievement of recommended physical activity levels.

‖Obesity is defined as body mass index ≥30 kg/m^2.

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further adjustment for other CVD risk factors (model 4), the association with MoCA was attenuated, while the association with DSST remained statistically significant. In sensitivity analyses excluding 136 participants with incident CVD during follow-up before cognitive testing, the results were materially unchanged (Figure S2). CAC progression showed a statistically significant interaction with race for MoCA (*P* for interaction=0.02), and the association tended to be stronger among Black participants (β=−0.13 [95% CI, −0.27 to 0.01]) than White participants (β, −0.01 [95% CI, −0.10 to 0.09]). A significant interaction by sex was also observed for MoCA (*P* for interaction=0.03), with the associations more pronounced among women (β, −0.17 [95% CI, −0.30 to −0.05]) than men (β, 0.01 [95% CI −0.10 to 0.12]). No statistically significant interactions were observed for the other cognitive outcomes.

DISCUSSION

We observed that baseline CAC >0 and CAC progression were prospectively associated with worse performance across several cognitive domains in midlife. Specifically, baseline CAC >0 was associated with lower processing speed, verbal memory, and global cognition, whereas CAC progression was associated with lower processing speed and global cognition in our primary model. Adjustment for CVD risk factors slightly attenuated these associations, but all associations remained statistically significant except for the association between CAC progression and global cognition. These associations persisted after excluding participants who developed CVD before cognitive testing, and the association with global cognition was more evident among Black than White participants and among women than men. Notably, the positive associations of CAC progression with worse cognitive performance remained after adjusting for baseline CAC score, suggesting that the association of CAC progression with worse cognitive performance may be independent of baseline CAC score. To our knowledge, this is the first study to examine the association between CAC progression and midlife cognitive performance in the general population.

The observed association between CAC progression and cognitive function in this study is consistent with a previous small study in the type 1 diabetes population.[7] In 148 middle-aged participants with childhood-onset type 1 diabetes, CAC progression, defined as the annualized difference between the square root of the baseline and the square root of the follow-up CAC score, was associated with cognitive impairment (odds ratio, 1.7 [95% CI, 1.1–2.9]) after adjustment

Table 2. Association Between Exposure to Coronary Artery Calcium Score and Cognitive Function in Midlife

	Baseline CAC score (year 15), mean (95% CIs)		Standardized cognitive differences (95% CIs)
	0 (n=2130)	>0 (n=211)	
Digit symbol substitution (processing speed)			
Model 1*	66.4 (65.8 to 67.1)	60.4 (58.2 to 62.6)	−0.37 (−0.51 to −0.23)
Model 2†	66.2 (65.6 to 66.8)	63.1 (61.1 to 65.0)	−0.19 (−0.32 to −0.06)
Model 3‡	66.3 (65.7 to 66.9)	63.0 (61.1 to 65.0)	−0.20 (−0.32 to −0.07)
Model 4§	66.2 (65.6 to 66.8)	63.8 (61.7 to 65.8)	−0.15 (−0.28 to −0.02)
Rey auditory verbal learning (verbal memory)			
Model 1*	8.7 (8.5 to 8.8)	7.4 (7.0 to 7.9)	−0.35 (−0.49 to −0.21)
Model 2†	8.6 (8.5 to 8.7)	7.9 (7.5 to 8.4)	−0.19 (−0.32 to −0.07)
Model 3‡	8.6 (8.5 to 8.7)	8.0 (7.6 to 8.4)	−0.18 (−0.31 to −0.06)
Model 4§	8.6 (8.5 to 8.7)	8.1 (7.7 to 8.6)	−0.14 (−0.27 to −0.01)
Stroop (executive function)			
Model 1*	−23.3 (−23.9 to −22.8)	−24.8 (−26.4 to −23.2)	−0.12 (−0.27 to 0.02)
Model 2†	−23.4 (−23.8 to −22.9)	−24.7 (−26.2 to −23.1)	−0.11 (−0.25 to 0.03)
Model 3‡	−23.3 (−23.8 to −22.9)	−24.7 (−26.3 to −23.2)	−0.12 (−0.25 to 0.02)
Model 4§	−23.4 (−23.9 to −22.9)	−24.0 (−25.6 to −22.4)	−0.05 (−0.19 to 0.09)
Montreal cognitive assessment (global cognition)			
Model 1*	24.2 (24.1 to 24.4)	23.2 (22.6 to 23.7)	−0.27 (−0.42 to −0.13)
Model 2†	24.2 (24.1 to 24.3)	23.4 (22.9 to 23.8)	−0.21 (−0.33 to −0.09)
Model 3‡	24.2 (24.1 to 24.4)	23.4 (22.9 to 23.8)	−0.21 (−0.33 to −0.09)
Model 4§	24.2 (24.1 to 24.3)	23.5 (23.1 to 24.0)	−0.17 (−0.29 to −0.04)
Fluency (verbal and category fluency)			
Model 1*	30.8 (30.4 to 31.1)	29.6 (28.5 to 30.8)	−0.14 (−0.29 to 0.00)
Model 2†	30.8 (30.4 to 31.1)	30.0 (28.9 to 31.1)	−0.09 (−0.23 to 0.04)
Model 3‡	30.8 (30.5 to 31.1)	29.9 (28.8 to 31.0)	−0.11 (−0.24 to 0.03)
Model 4§	30.8 (30.5 to 31.1)	30.2 (29.1 to 31.3)	−0.08 (−0.21 to 0.06)

Data are estimated marginal mean and standardized cognitive differences (95% CIs). Digit symbol substitution test (range, 0–133); Rey auditory verbal learning test (range, 0–15); Stroop test (original range, 0–160; multiplied by −1 to yield a −160 to 0 scale, where higher scores indicate better performance); Montreal cognitive assessment (range, 0–30); fluency (range, ≥0). Higher scores indicate better cognitive performance for all cognitive tests. CAC indicates coronary artery calcium.

*Model 1 was age-adjusted.

†Model 2 additionally adjusted for sex, race, and education.

‡Model 3 additionally adjusted for physical activity, depressive symptoms, and APOE.

§Model 4 additionally adjusted for hypertension, diabetes, obesity, dyslipidemia, current smoking, and current drinking.

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for baseline CAC score. MESA, which included 4173 participants with a mean age of 64.0 years, reported a higher risk of incident dementia and a lower mean Cognitive Abilities Screening Instrument score among those with CAC progression, but it did not reach statistical significance after adjusting for baseline CAC score.[8] Possible explanations for the association observed in the present study may be the younger age of the target population (40.2 years at baseline in the present study) compared with the mean age of 64.0 years in MESA. Furthermore, MESA performed repeat CAC scans over ≈2.5 years, which may not have been sufficient to capture the CAC progression over time. The present study defined CAC progression based on 3 CAC measurements taken at 5-year intervals over ≈10 years, resulting in a more comprehensive assessment of the dynamic nature of CAC.

The mechanisms by which CAC progression affects cognitive function are unclear. Coronary calcified plaque has been associated with greater central nervous system vascular damage as assessed with white matter lesion volume, white matter microstructural integrity deterioration, cerebral infarcts, and cerebral microbleeds.[4] Chronic cerebral hypoperfusion due to cardiac dysfunction may also adversely impact cognitive function. Neuropathological studies suggest that exposure to CVD risk factors might also disrupt amyloid clearance and production and increase amyloid deposition and plaques.[27,28] Exposure to CVD risk factors may lead to neuronal damage due to increased inflammation and oxidative stress.[29–31]

The strengths of the present study include its longitudinal design with the sequential CAC scans and collection of comprehensive data on cognitive function and potential confounders. However, our study has several limitations.

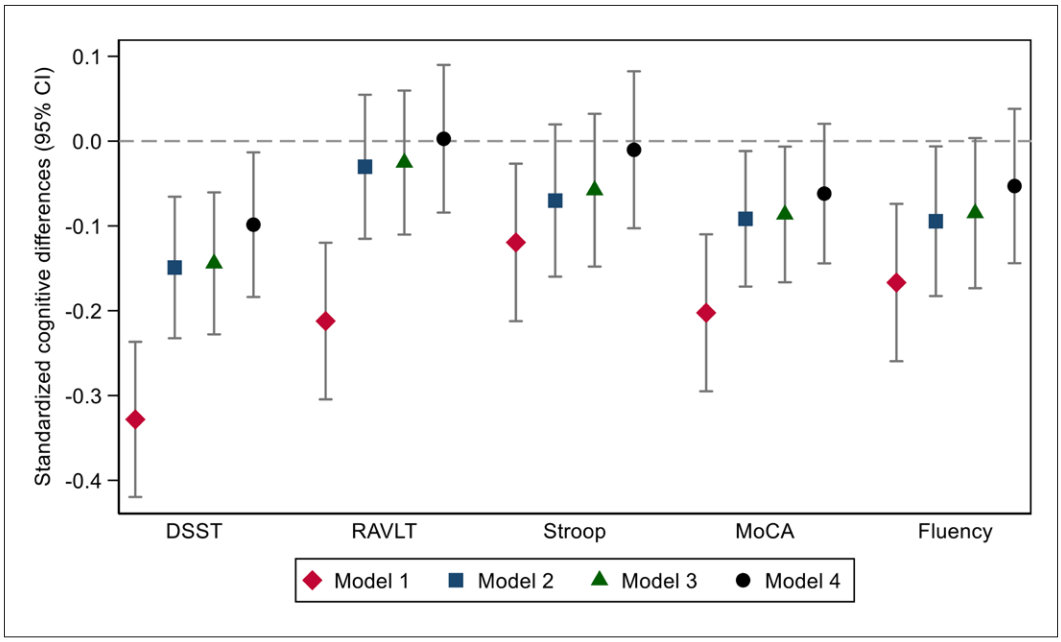

Figure. Association between exposure to coronary artery calcium progression and cognitive function in midlife. Standardized cognitive differences (95% CI) for coronary artery calcium progression vs no progression for the 5 tests. Model 1 was age-adjusted; model 2 additionally adjusted for sex, race, education, and baseline coronary artery calcium; model 3 additionally adjusted for physical activity, depressive symptoms, and APOE; and model 4 additionally adjusted for traditional cardiovascular risk factors (hypertension, diabetes, obesity, dyslipidemia, current smoking, and current drinking). DSST indicates digit symbol substitution test (processing speed); Fluency, average of letter and category fluency tests (verbal and category fluency); MoCA, Montreal cognitive assessment (global cognition); RAVLT, Rey auditory verbal learning test (verbal memory); and Stroop, Stroop Test (executive function).

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First, although we adjusted for multiple potential confounders, including demographic, clinical, and genetic factors, unmeasured social, structural, and environmental factors, and residual confounding may still bias the observed association. Second, the clinical significance of modest differences in cognitive performance between the CAC score and progression groups is unclear. In the present study, the largest difference was observed in the DSST. Since a lower DSST score may predict future clinical disorders of cognition, mobility, and mood,[32] the difference in cognitive performance observed in this study would show early cognitive changes and extend to later impairments of other cognitive domains. Further studies are warranted to determine whether the association of subclinical atherosclerosis with worse cognitive performance in midlife leads to cognitive decline and dementia in older age. Finally, the generalizability of our findings to other racial/ethnic populations may be limited because our cohort only included Black and White adults.

In conclusion, the present study showed that CAC score and progression were associated with worse performance across several cognitive domains in midlife, particularly processing speed and global cognition. The associations of CAC progression with worse cognitive performance remained after adjustment for baseline CAC score, suggesting that CAC progression could be a risk indicator for early cognitive aging, independent of baseline CAC score and other risk factors. Assessment of CAC progression through repeated CAC scans may provide additional clinical value, beyond baseline CAC alone, for identifying individuals at increased risk for midlife cognitive decline. Further studies are needed to examine whether timely identification and management of subclinical atherosclerosis prevents early cognitive aging.

ARTICLE INFORMATION

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Author Contributions

Dr Teramoto is responsible for study conceptualization, data analysis, and drafting of the article. J.G. Terry, Dr Sidney, and Dr Yaffe are responsible for data acquisition. J.G. Terry, Dr Pletcher, Dr Marcus, Dr Hoffmann, Dr Sidney, and Dr Yaffe are responsible for intellectual input and revision of the article. All authors approved the final version.

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Disclosures

None.

Supplemental Material

Table S1
Figures S1 and S2

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